
# *Assignment 1 & 2*

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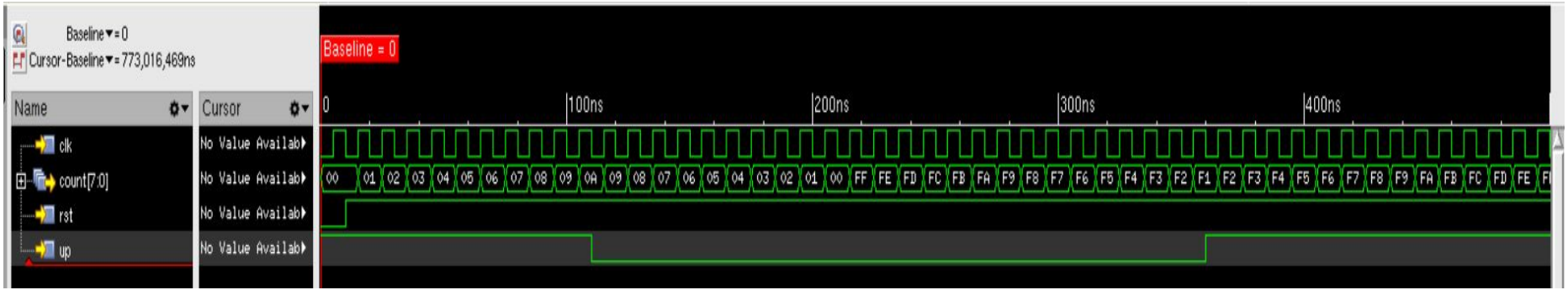
**ASIC design**

# Part 1: RTL Simulation



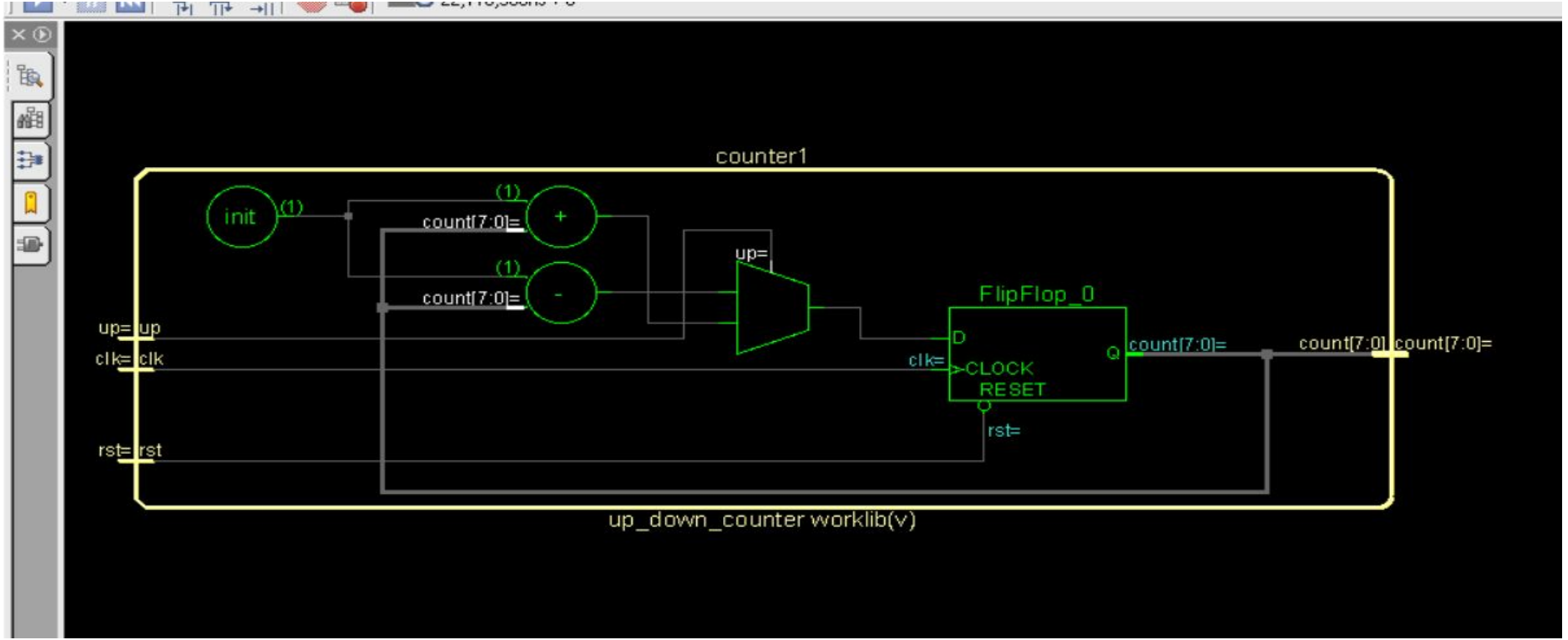

By adding an up input bit we can control whether the counter increments or decrements , we can see that when up=1 the counter increments , when it goes low , the counter starts decrementing until 00 and overflows to FF , by pulling up back again the counter goes up from F1,F2…

# RTL and Testbench

```
module up_down_counter(clk,rst,up,count);
input clk,rst,up;
output reg [7:0] count;
always@(posedge clk or negedge rst)
begin
if(!rst)
count<=0;
else
begin
if(up)
count<=count+1;
else
count<=count-1;
end
end
endmodule
```

```
module counter_test;
reg clk, rst ,up;
wire [7:0] count;
initial
begin
clk=0;
rst=0;
up=1;
#10;
rst=1;
#100;
up=0;
#250;
up=1;
end
up_down_counter counter1(clk,rst,up, count);
always #5 clk=~clk;
endmodule
```

Schematic view of the design.



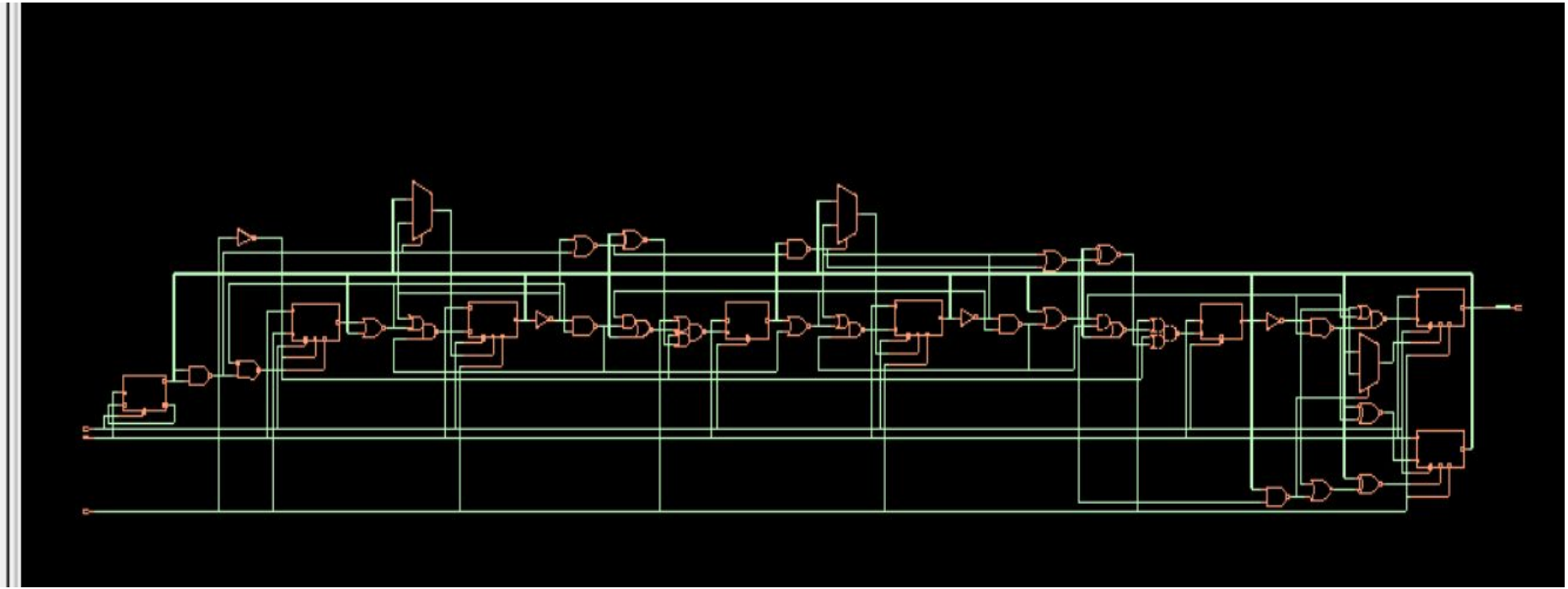

The RTL simulation verifies the *functional correctness* of the up-down counter. At this stage, the design is described behaviorally using arithmetic and control constructs such as `+`, `-`, and conditional statements. The simulator models these operations abstractly

# Timing Constraints

```
create_clock -name clk -period 10 -waveform {0 5} [get_ports "clk"]
set_clock_transition -rise 0.1 [get_clocks "clk"]
set_clock_transition -fall 0.1 [get_clocks "clk"]
set_clock_uncertainty 0.01 [get_ports "clk"]
set_input_delay -max 1.0 [get_ports "rst"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "count"] -clock [get_clocks "clk'
set_input_delay -max 1.0 [get_ports "up"] -clock [get_clocks "clk"]
```

A constraint on “up” input was added for Part II:Synthesis

# Part II:Synthesis



After synthesis , the RTL code is mapped to *standard cells* from the technology library. The resulting gate-level netlist replaces high-level arithmetic operations with combinational logic made of AND, OR, XOR, and MUX gates, along with D-flip-flops for storage.
The synthesized circuit represents the *actual hardware implementation* that would be realized on silicon

# Screenshot of the generated netlist

```
module up_down_counter(clk, rst, up, count);
  input clk, rst, up;
  output [7:0] count;
  wire clk, rst, up;
  wire [7:0] count;
  wire n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7;
  wire n_8, n_9, n_10, n_11, n_12, n_13, n_14, n_15;
  wire n_16, n_17, n_18, n_19, n_20, n_21, n_22, n_23;
  wire n_24, n_25, n_26, n_27, n_28, n_29, n_30, n_31;
  SDFFRHQX1 \count_reg[7] (.RN (rst), .CK (clk), .D (n_31), .SI (n_29),
       .SE (up), .Q (count[7]));
  SDFFRHQX1 \count_reg[6] (.RN (rst), .CK (clk), .D (n_28), .SI (n_26),
       .SE (up), .Q (count[6]));
  DFFRHQX1 \count_reg[5] (.RN (rst), .CK (clk), .D (n_30), .Q
       (count[5]));
  XNOR2X1 g676__2398(.A (count[7]), .B (n_27), .Y (n_31));
  OAI22X1 g681__5107(.A0 (up), .A1 (n_21), .B0 (n_19), .B1 (n_18), .Y
       (n_30));
  XNOR2X1 g677__6260(.A (count[7]), .B (n_23), .Y (n_29));
  SDFFRHQX1 \count_reg[4] (.RN (rst), .CK (clk), .D (n_17), .SI (n_14),
       .SE (up), .Q (count[4]));
  OAI21X1 g679__4319(.A0 (n_25), .A1 (n_24), .B0 (n_27), .Y (n_28));
  DFFRHQX1 \count_reg[3] (.RN (rst), .CK (clk), .D (n_20), .Q
       (count[3]));
  MXI2XL g680__8428(.A (count[6]), .B (n_25), .S0 (n_22), .Y (n_26));
  NAND2X1 g682__5526(.A (n_25), .B (n_24), .Y (n_27));
  OR2XL g683__6783(.A (n_25), .B (n_22), .Y (n_23));
  AOI21X1 g685__3680(.A0 (count[5]), .A1 (n_16), .B0 (n_24), .Y (n_21));
  OAI22X1 g692__1617(.A0 (up), .A1 (n_11), .B0 (n_19), .B1 (n_9), .Y
       (n_20));
  XNOR2X1 g686__2802(.A (count[5]), .B (n_15), .Y (n_18));
  OAI21X1 g690__1705(.A0 (n_13), .A1 (n_12), .B0 (n_16), .Y (n_17));
  SDFFRHQX1 \count_reg[2] (.RN (rst), .CK (clk), .D (n_7), .SI (n_5),
       .SE (up), .Q (count[2]));
  NAND2X1 g688__5122(.A (count[5]), .B (n_15), .Y (n_22));
  NOR2X1 g687__8246(.A (count[5]), .B (n_16), .Y (n_24));
  MXI2XL g691__7098(.A (count[4]), .B (n_13), .S0 (n_10), .Y (n_14));
  SDFFRHQX1 \count_reg[1] (.RN (rst), .CK (clk), .D (up), .SI (n_19),
       .SE (n_2), .Q (count[1]));
  NAND2X1 g693__6131(.A (n_13), .B (n_12), .Y (n_16));
  AOI21X1 g696__1881(.A0 (count[3]), .A1 (n_8), .B0 (n_12), .Y (n_11));
  NOR2X1 g694__5115(.A (n_13), .B (n_10), .Y (n_15));
  XNOR2X1 g697__7482(.A (count[3]), .B (n_6), .Y (n_9));
  NOR2X1 g699__4733(.A (count[3]), .B (n_8), .Y (n_12));
  OAI21X1 g701__6161(.A0 (n_4), .A1 (n_1), .B0 (n_8), .Y (n_7));
```

```
NAND2X1 g700__9315(.A (count[3]), .B (n_6), .Y (n_10));
MXI2XL g702__9945(.A (count[2]), .B (n_4), .S0 (n_3), .Y (n_5));
NOR2X1 g705__2883(.A (n_4), .B (n_3), .Y (n_6));
NAND2BX1 g704__2346(.AN (n_1), .B (n_3), .Y (n_2));
NAND2X1 g703__1666(.A (n_4), .B (n_1), .Y (n_8));
NOR2X1 g707__7410(.A (count[1]), .B (count[0]), .Y (n_1));
NAND2X1 g708__6417(.A (count[1]), .B (count[0]), .Y (n_3));
INVX1 g711(.A (count[4]), .Y (n_13));
INVX1 g712(.A (count[2]), .Y (n_4));
INVX1 g713(.A (up), .Y (n_19));
INVX1 g710(.A (count[6]), .Y (n_25));
DFFRX1 \count_reg[0] (.RN (rst), .CK (clk), .D (n_0), .Q (count[0]),
     .QN (n_0));
endmodule
```

# Timing Report

```
====================================================================
  Generated by:           Genus(TM) Synthesis Solution 22.14-s059_1
  Generated on:           Oct 08 2025  01:49:47 pm
  Module:                 up_down_counter
  Operating conditions:   PVT_0P9V_125C (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
====================================================================


Path 1: MET (8677 ps) Late External Delay Assertion at pin count[0]
         Group: clk
    Startpoint: (R) count_reg[0]/CK
         Clock: (R) clk
      Endpoint: (R) count[0]
         Clock: (R) clk

                   Capture       Launch
       Clock Edge:+   10000            0
       Src Latency:+      0            0
       Net Latency:+      0 (I)        0 (I)
           Arrival:=  10000            0

     Output Delay:-    1000
    Required Time:=    9000
     Launch Clock:-       0
        Data Path:-     323
            Slack:=    8677

Exceptions/Constraints:
  output_delay             1000            constraints_top.sdc_line_6_7_1

#-------------------------------------------------------------------------------------------
#  Timing Point   Flags   Arc   Edge   Cell     Fanout Load Trans Delay Arrival Instance
#                                                (fF)  (ps)  (ps)   (ps)  Location
#-------------------------------------------------------------------------------------------
  count_reg[0]/CK -       -     R     (arrival)      8    -   100     0       0    (-,-)
  count_reg[0]/Q  -       CK->Q R     DFFRX1         3  0.8    36   323     323    (-,-)
  count[0]        <<<     -     R     (port)         -    -     -     0     323    (-,-)
#-------------------------------------------------------------------------------------------
```

The up-down counter shows a very large positive slack of 8677 ns, which indicates that the circuit is extremely fast compared to the specified 10 ns clock period. This occurs because the design is small and simple — consisting of only basic logic (a few gates and flip-flops) — resulting in very short data path delays Thus, we can increase the clock frequency (reduce the clock period) to better utilize the circuit’s speed

# Power Report

```
Instance: /up_down_counter
Power Unit: W
PDB Frames: /stim#0/frame#0
  ------------------------------------------------------------------------
    Category        Leakage     Internal    Switching        Total    Row%
  ------------------------------------------------------------------------
      memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
    register     1.28043e-09  2.58607e-06  4.64638e-08  2.63381e-06  83.19%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       logic     1.03037e-09  2.83765e-07  1.17834e-07  4.02629e-07  12.72%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       clock     0.00000e+00  0.00000e+00  1.29600e-07  1.29600e-07   4.09%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
  ------------------------------------------------------------------------
    Subtotal     2.31080e-09  2.86983e-06  2.93898e-07  3.16604e-06 100.00%
  Percentage           0.07%       90.64%        9.28%      100.00% 100.00%
  ------------------------------------------------------------------------
```

The synthesized *up-down counter* shows that switching power contributes about 9.28% of the total power, while leakage accounts for only 0.07%, with the remainder coming from internal power.
The clock network consumes around 4.09% of the total power, which is typical for a small sequential circuit.
Most of the power is spent in the combinational logic (adders and multiplexers) due to signal transitions.

# Area Report

```
====================================================================
  Generated by:           Genus(TM) Synthesis Solution 22.14-s059_1
  Generated on:           Oct 08 2025  01:50:26 pm
  Module:                 up_down_counter
  Operating conditions:   PVT_0P9V_125C (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
====================================================================

    Instance    Module  Cell Count  Cell Area  Net Area   Total Area   Wireload
--------------------------------------------------------------------------------
up_down_counter                 39    106.362     0.000      106.362 <none> (D)
  (D) = wireload is default in technology library
```

The synthesized *up-down counter* consists of 39 cells, occupying a total cell area of 106.362. No additional area is reported for nets (net area = 0) since no wireload model was applied.

# qor Report

```
  Generated on:            Oct 08 2025  01:50:46 pm
  Module:                  up_down_counter
  Operating conditions:    PVT_0P9V_125C (balanced_tree)
  Wireload mode:           enclosed
  Area mode:               timing library
================================================================

Timing
--------

Clock  Period
--------------
clk   10000.0


  Cost    Critical         Violating
 Group   Path Slack  TNS     Paths
---------------------------------------
clk          8677.4   0.0         0
default    No paths   0.0
---------------------------------------
Total                 0.0         0

Instance Count
--------------
Leaf Instance Count              39
Physical Instance count           0
Sequential Instance Count         8
Combinational Instance Count     31
Hierarchical Instance Count       0

Area
----
Cell Area                         106.362
Physical Cell Area                0.000
Total Cell Area (Cell+Physical)   106.362
Net Area                          0.000
Total Area (Cell+Physical+Net)    106.362

Max Fanout                        8 (clk)
Min Fanout                        0 (rst)
Average Fanout                    2.4
Terms to net ratio                3.3488
Terms to instance ratio           3.6923
Runtime                           53.779793 seconds
Elapsed Runtime                   903 seconds
Genus peak memory usage           1807.07
Innovus peak memory usage         no_value
```

The synthesized counter has a maximum fanout of 8, minimum of 0, and an average fanout of 2.4, indicating moderate loading on most signals.
The terms-to-net ratio of 3.35 and terms-to-instance ratio of 3.69 suggest that nets typically connect a small number of pins relative to the number of cells, reflecting the simple structure of the design.

# Screenshot of generated reports directory.